



Benoît COURBIN

Parametric Design, Naval, Interior Architecture and A.I. Tools

🏠 Barcelona, Spain

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Transdisciplinary designer with 20 years of experience in design offices, architecture and specialized sectors (textiles, naval). I design and optimize projects through parametric design, advanced digital tools and artificial intelligence applied at AECO, a field in which I am currently doing a master's degree. Teacher for more than ten years (Rhino 3D, Grasshopper, V-Ray), I now put my expertise at the service of companies looking for a creative and innovative profile, capable of linking design, emerging technologies and strategic vision.

Languages

French: native
Spanish: current (C1)
English: Conversational (B2)
Catalan: Beginner (A2)

Personal Skills

Teamwork Time
Management
Problem Solving
Autonomy
Creativity

Skills

3D Modeling + Rendering
Computer Graphics
Plan File Creation
Project Management
Site Monitoring
Quotes and Cost Calculation
Technical Reports
Implementation of AI tools
Teaching

Software

2D : AutoCAD, PowerCad.
3D : Rhinoceros + Grasshopper,
SketchUp, 3ds Max.
BIM : Revit + Dynamo, Archicad
Mechanical: SolidEdge, SolidWorks
Rendering: V-Ray, KeyShot.
Computer graphics: Photoshop,
Illustrator, InDesign, Acrobat.
Pack Office (+ VBA).
Hydrostatic: Circée Maat.
A.I.: n8n, midjourney, etc.

Professional Experiences

• Freelance draftsman in interior design

Design and follow-up of renovations.
Creation of images and plans for residential projects (single-family houses, collective housing) and tertiary projects (offices, restaurants).

• Teacher in 3D Modeling and Rendering

École CYDesign – Université Paris-Cergy – Saint-Germain-en-Laye, France – 2023
<https://www.cy-ecolededesign.fr/>

Modeling with Rhinoceros 3D.
Volumetric, surface and parametric.
Renderings (objects, architectures).
Infographics / plans.
Preparing files for production.

• Head of the integrated design office in Textile Architecture – 3 years

AB2CS (textile architecture and circus tent) – Bordeaux (France) – 2012-2015

Design of textile membranes and mechanical parts.
Double curvature development.
Technical watch and evolution of standards. Prototype development.
Follow-up of installations.
Manufacture of circus tents.
International collaboration.
Creation of specific software.

• Teacher of 3D modeling and digital tools – 6 years

École Camondo (Les Arts Décoratifs) – Paris (France) – 2009-2015
<https://www.ecolecamondo.fr/en/>

• Architectural Draftsman – and 3D Graphic Designer – 3 years

Paranteau-Ithurbide Architecture Agency – Bordeaux (France) – 2006-2009

Research (conceptual and formal)
Timber frame, tertiary and industrial building
2D, 3D computer graphics
Creation of plan folders.
Follow-up of the realization of interior fittings.

- **Naval Architect** - 6 months

Naval architecture firm Chenus Presles et Associés (C.P.A.) - Paris (France) - 2005

Hull design

Business Plans - 3D Renderings

Basis in Stability Calculation and

Sampling

Mass Quote

- **Naval Designer** - 3 years

Couach shipyard _ Gujan Mestras (France) - 2001- 2004

<https://couach.com/fr/>

Business Plans - 3D Renderings

Design of Interior Fittings

Deck plans, choice Fittings

Design of specific parts (hoods, mold insert, etc.)

Method and production

- **Industrial Designer** - 7 months

Spie S.O. (fibre optic network and street lighting)_Bordeaux (France)

Implementation of the management of public lighting in the city of Arcachon.

Deployment of fibre optics on the Gironde.

Formation

Master's Degree in Artificial Intelligence for AECO

Zigurat Global Technology Institute – Ongoing University of Barcelona (2026)

<https://www.e-zigurat.com/es/master-artificial-intelligence-architecture-construction/>

C.E.A.A. Naval Architecture

École Nationale Supérieure d'Architecture de Paris–La Villette – 2005

<https://paris-lavillette.archi.fr/formations/post-masters/dpea-architecture-navale/>

D.E.S.C.D. Product Design section

I.E.D. Toulon – 2003

Alternating with the Couach shipyards

D.U.T. Mechanical and Production Engineering

University of Bordeaux 1 – 2001

<https://www.iut.u-bordeaux.fr/gmp/>

More about me

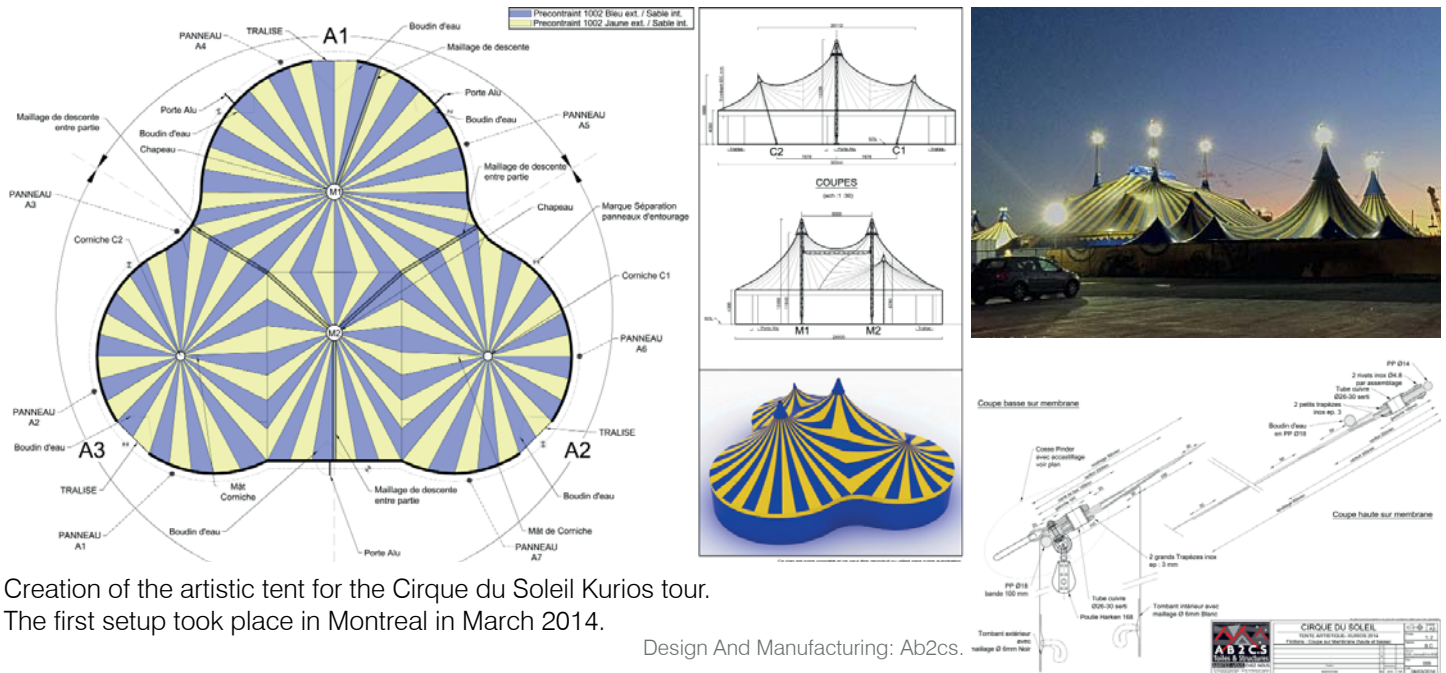
I am a native French speaker and father of two, living in Barcelona for the past ten years. Curious and open-minded, I enjoy discovering new cultures through travel, encounters, and the arts.

Passionate about music, I play the guitar, and I also love sailing, chess, comics, architecture, design, and graphic arts. In my free time, I explore Barcelona with my children or meet friends to share good moments and new inspirations.



Tensile Architecture





La petite Bambou



Blue Ocean

Roofing of the terrace at the Blue Océan campsite in Ondres.

Design And Manufacturing: Ab2cs.



Cité des Pyrénées

Cladding of the exterior windows of the Cité des Pyrénées in Pau

Design And Manufacturing: Ab2cs.



Roof covering for a restaurant terrace with an impluvium in Montreal (Canada).

Manufacturing: Ab2cs.



Olympic Stadium of Montreal

AB2CS-2014

Manufacturing of the partitions between the gym, the dojo, and the pool during the renovation project of the Olympic Stadium in Montreal (Canada).

Manufacturing: Ab2cs

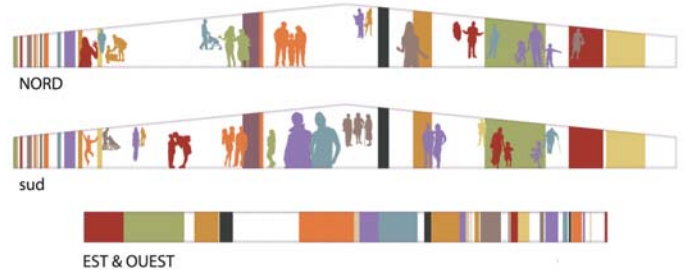


Halles de Bégard

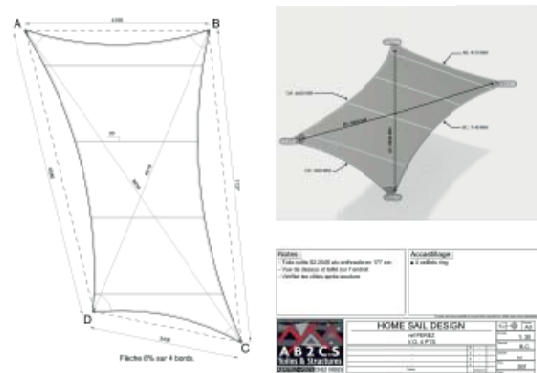
AB2CS-2014

Creation of textile façades with digital printing at the Halles de Bégard.

Design, manufacturing & graphic design: AB2CS. **



Shade sails



Example of a shade-sail plan created using the program developed in Grasshopper, optimizing design time, raw materials, and the cost of fabric corners by standardizing them.

Pavilion M

AB2CS-2012

Pavilion M is the welcome center for the events surrounding Marseille 2013 – European Capital of Culture. It is a temporary building measuring 100 meters long and 15 meters wide, made from a tubular grid structure covered with CTBX.

The manufactured membrane covers the building, serving both as the roof and the interior ceiling.

The main challenge of this project was the study of the membrane tensioning configuration..

Design & Manufacturing: AB2CS

Project Management: GL Event

Engineering: Abaca

Metal Structure: Mash



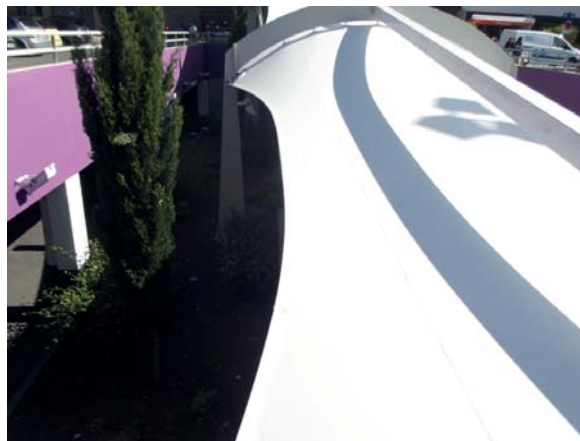
La Valentine

AB2CS-2013



This project, located in the La Valentine shopping center in eastern Marseille, involves the replacement of membranes on a metal-textile structure built in 1999. We manufactured the membranes taking into account the defects observed in the old membranes (issues with tensioning and the formation of water pockets).

Manufacturing : AB2CS



Residence Cap Roux

AB2CS-2013



Cladding of the balconies and windows of the two buildings of the Cap Roux Residence in Mérignac. The balcony membranes, made of FT 381 (Ferrari fabric), are tensioned over PTS aluminum profiles. The geometries of the balconies and the parapets above the windows required the membranes to be tensioned in curved shapes to respect the curvatures desired by the architect.

Client: Gironde Habitat
Design and Manufacturing: AB2CS

CinéMarivaux in Macon

AB2CS-2014



Cladding of 1,250 m² of cinema façades in 16 modules. Supply and installation of the secondary structure that allows the membranes to be tensioned. Oversight of the digitally printed membranes carried out by sub-contractors.

Client: CINÉMA IVAUX
Design: CERIS
Manufacturing: AB2CS

Public Finance Center



Manufacturing of the shade sails and textile façades for the new Public Finance Center on Réunion Island. The main challenge of this project was ensuring resistance to the required forces: withstand winds of over 260 km/h!

Design and Manufacturing: AB2CS

Pia Huvaune

Manufacturing of the canvas umbrellas for 3 residential buildings in the renovation project of the Vélodrome Stadium in Marseille. Clad surface:

Building A: 300 m²
Building B: 330 m²
Building C: 470 m²

Design: ASTEO
Structure: SIRC
Manufacturing: AB2CS

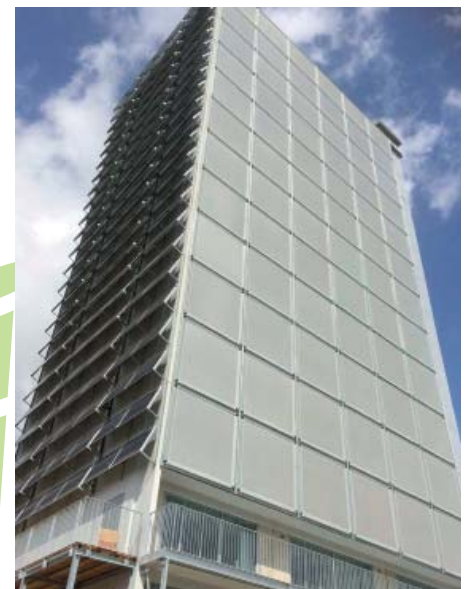
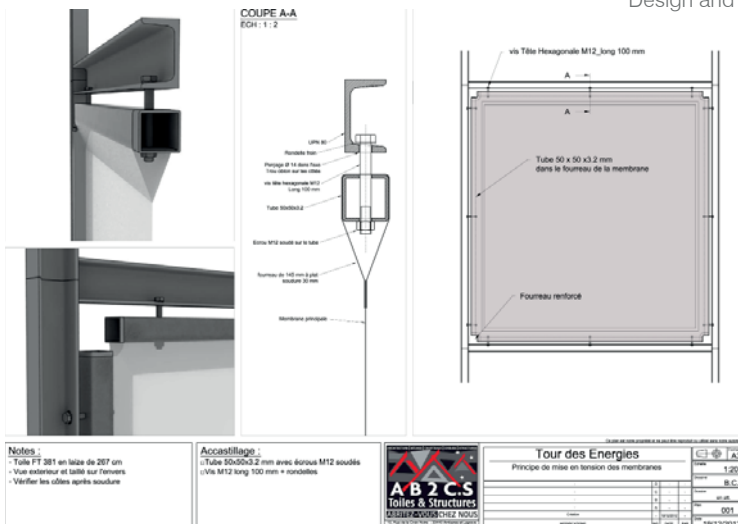


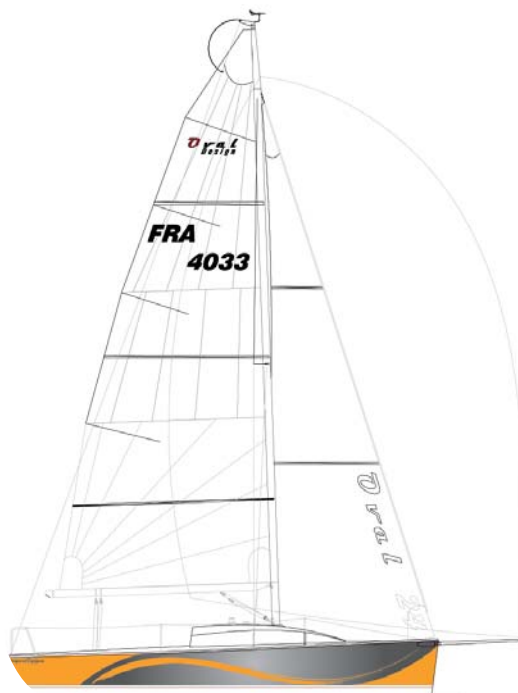
Energy Tower in Nancy

AB2CS-2013

Cladding of two façades during the tower's renovation. The tensioning of the 144 modules is achieved using tubes. Membranes: FT 381.

Design and Manufacturing: AB2CS





Naval Architecture



Eco

DPEA Project – 2005



Specifications

Waterline length: 7.65 m
 Hull beam: 2.55 m
 Waterline beam: 2 m
 Lightweight: 1.825 t
 Full load weight: 2.4 t
 Sail area (pressure): 35 m²
 Draft: 1.75 / 0.9 m
 Keel: 600 kg
 Engine: 6 HP outboard
 Design category: B



This project originated from the eco-marine competition launched by Loisirs Nautiques magazine in autumn 2004. The brief called for the design of a boat no longer than 12 meters, with non-mechanical propulsion, trailerable or not, taking into account environmental impact during both construction and use.

Eco is an 8-meter coastal trailerable boat with a beam of only 2.55 meters (compliant with road size regulations). Designed for amateur construction, its goal is to reduce manufacturing costs (no molds) and environmental impact. It is therefore built from plywood reinforced with composites, depending on specific areas (keel trunk, bow, frames).

The interior includes a forward area with a double bed, a convertible salon with two beds, a kitchenette, a single bed under the helm, and a wardrobe.

Duc d'Albe

CPA - 2003

"Duc d'Albe" is a river-sea steel coastal vessel offering nearly 50 m² of living space distributed over two and a half levels, with a cruising speed of 7 knots. Its interior includes two bedrooms, two bathrooms, a living room, a helm-lounge-kitchen area, and an upholstery workshop. The exterior features a terrace and a cockpit..



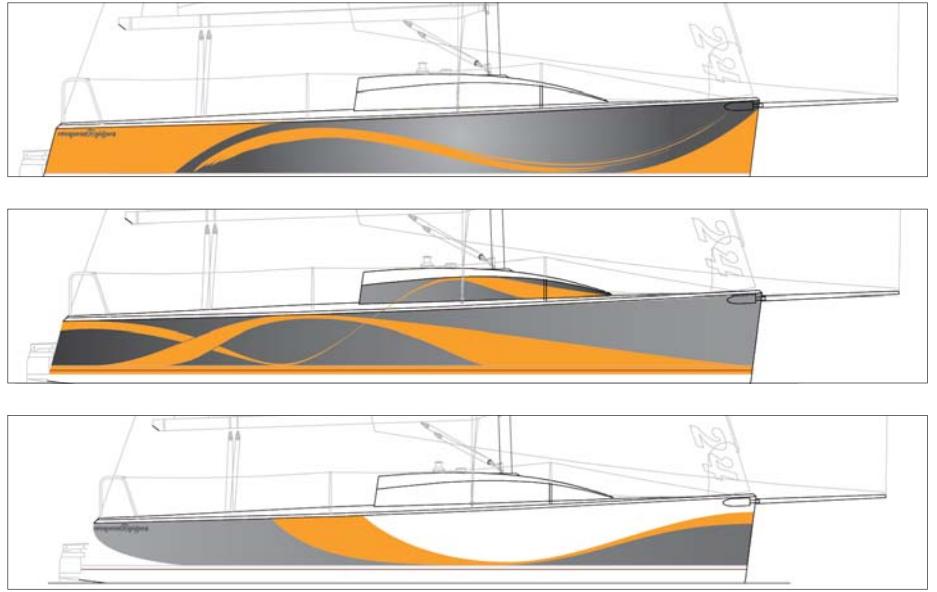
Specifications:

Overall length: 14.95 m
 Overall beam: 4.36 m
 Displacement: 28 t
 Draft: 1.00 m
 Air draft: 3.45 m
 Engine: 120 HP inboard

Arquitect : C.Chenus

Croqueur 2 Pignes

Freelance Project-2008



Croqueur 2 Pignes is a 24-foot (approximately 7.30 m) racing boat prototype, built in the Arcachon basin. In this project, I participated in the custom (one-off) construction and oversaw the decoration. Here are some of the proposals created. Launched in spring 2009.

Consola de Flybridge

Couach - 2003



These control panels were designed for the Guy Couach shipyard during the 2003 fleet renovation. On this occasion, the shipyard expressed the desire to design a control panel intended to equip the flybridges of models ranging from 22 to 28 meters.

The shipyard asked me to make proposals considering several constraints: standardization of molds, reduction of assembly parts, respect for dimensions, consideration of the flybridge layout, versatility in equipment placement (variable in number and model according to client choices), and improved ergonomics.

The panels are composed of two laminated parts: the base and the dashboard.

The base houses the electrical and hydraulic relays and allows movement of the drop cover, mounted on sliders.

The dashboard consists of a horizontal plane (for installation of the throttle levers and main controls), three vertical planes (for optimal reading of electronic instrumentation), and a visor. Finally, the hydraulic steering pump is integrated within the thickness of the horizontal plane.



Product Design



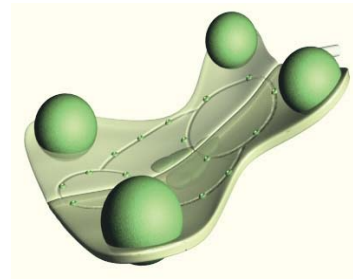
BAZAO

DESCD Project - 2003

The Bazao project is a workshop carried out by IED in collaboration with the pool manufacturer Bleu Azur.

Bazao is a pool base that integrates filtration, lighting, a counter-current swimming system, and a diving board, to which various equipment can be connected, such as a cleaning robot, a fountain, a relaxation mat, water games, and a ladder.

With its organic lines and carefully chosen colors, this installation immerses itself in a playful and child-friendly universe, designed to move the water and provide maximum enjoyment for both children and adults.



Wood Stove

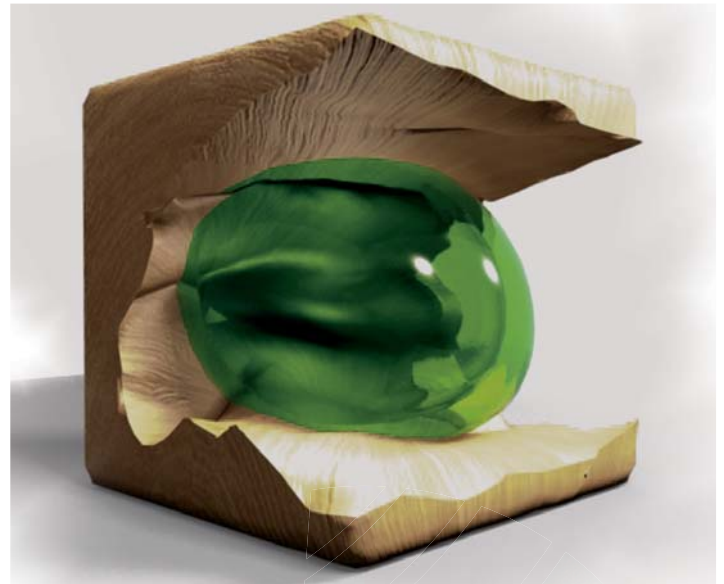
Freelance Project for Félix & associés design - 2010.

This project was developed for a company in Périgord specializing in the folding and assembly of metal sheets. It originated from the filing of a patent related to heating the domestic water network using a wood stove. The entire stove assembly is manufactured using the company's production facilities, except for the cast iron façade.



Format Raisin

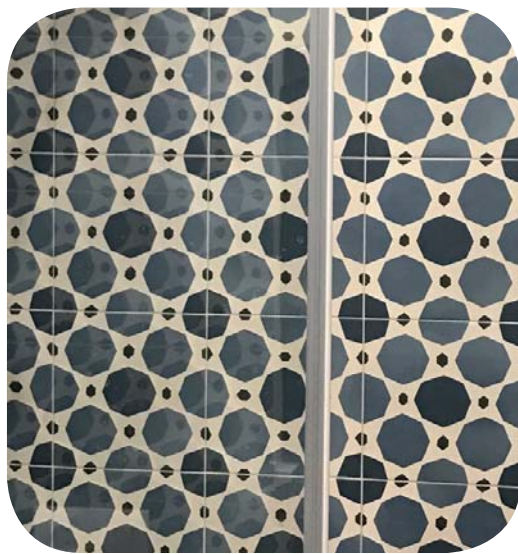
DESCD Project - 2003



Les Bodegas du Var organized a design competition with the aim of creating a trophy to reward innovations in wine marketing.

This trophy symbolizes "the idea," represented by the color and transparency of glass, carefully protected in a solid wood box, hollowed on three sides.

The materials - glass and wood - reflect the language of wine, while the shapes - a carved wooden cube and an ovoid blown-glass form - evoke craftsmanship and the human touch, essential elements in wine production.



Architecture
& Interior Design



Casa de ciudad en Issy-les-Moulineaux

Freelance Project - 2006



This project was carried out for an S.C.I. in Issy-les-Moulineaux.

The pavilion, with a living area of 75 m², included a ground floor, a split-level basement, a room above the basement, a first floor, a mezzanine intended for storage, and an interior garden.

Designed to become a shared home for three people, the program was defined with the following main requirements: a large living room, three bedrooms, a comfortable kitchen, at least two bathrooms, and a separate WC. Private spaces were located on the upper levels, while the split-level basement was transformed into a kitchen overlooking the garden. The ground floor was left open-plan, creating a spacious living area for the three housemates.

Bathroom Renovation

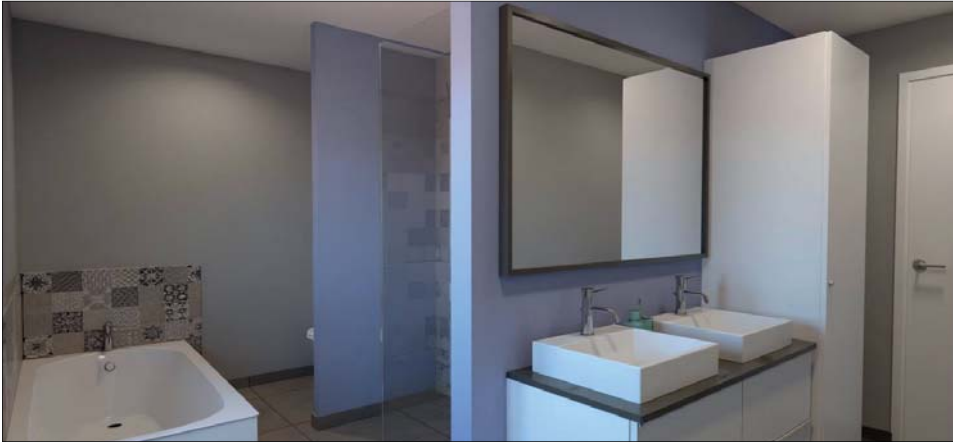
Agencia Parenteau-Ithurbide - 2008



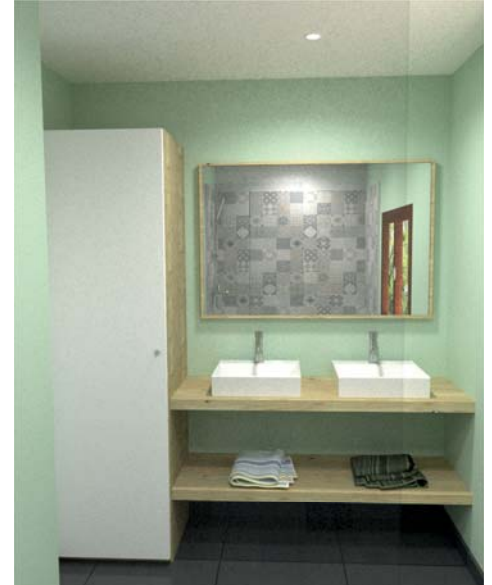
Project carried out in Haillan, near Bordeaux
3D images of the selected solution.

Renovation of 2 bathrooms

projet freelance - 2006



Images of the approved proposals for the renovation of two bathrooms and an office in a villa in Martinique..



Renovation of a villa in Les Trois-Îlets

Freelance - 2020

Images and plans of the transformation project of a villa in Les Trois-Îlets.

Main modifications:

- Creation of a swimming pool and a terrace
- Construction of an access ramp
- Renovation of the kitchen and creation of an office space



Preliminary design of a take-away restaurant

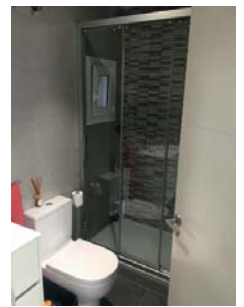
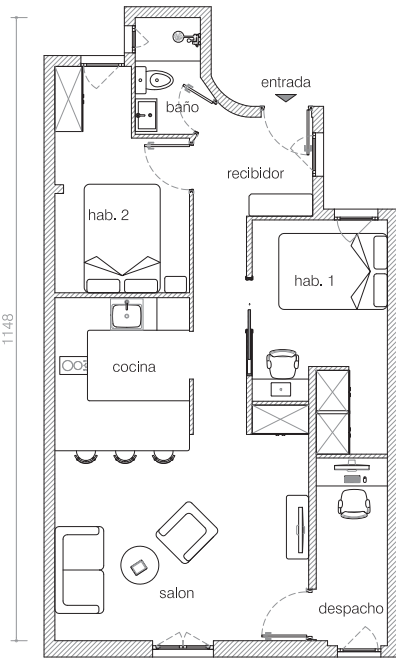
Freelance - 2021

Design of the visual identity, façade, and interior architecture of a fast-food restaurant in Fort-de-France



Renovation of a 62 m² apartment

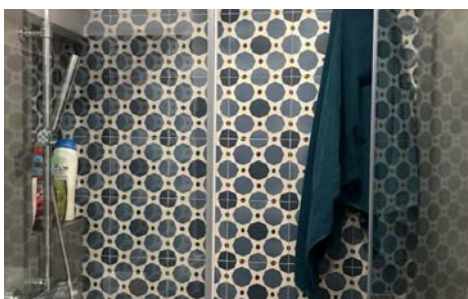
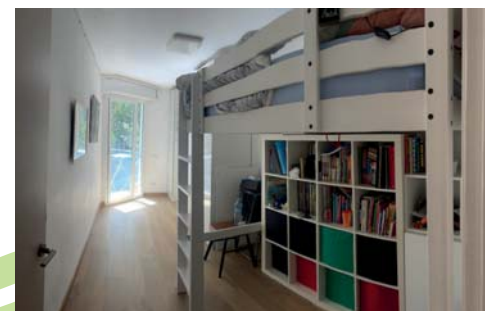
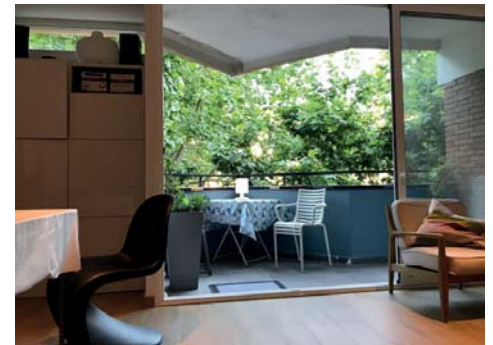
Freelance- 2019



Complete renovation of a 62 m² apartment in the Born district of Barcelona, comprising 2 double bedrooms, an open kitchen, and a study.

Renovation of a 106 m² apartment

Freelance - 2018



Complete renovation of a 106 m² duplex apartment in the Ciutadella district of Barcelona, comprising a master suite, 2 single bedrooms, a laundry room, 2 bathrooms, and a living area open to the terrace and surrounding greenery.

Creation of a hotel in the Bordeaux region

Parenteau Ithurbide Agency
2008



Image accompanying the Preliminary Design (Concept) of a hotel near Bordeaux-Mérignac Airport.

Haut-Lormont Residence

Parenteau Ithurbide Agency
2007



Image accompanying the Preliminary Design (Concept) of a senior residence on the outskirts of Bordeaux (Haut-Lormont).

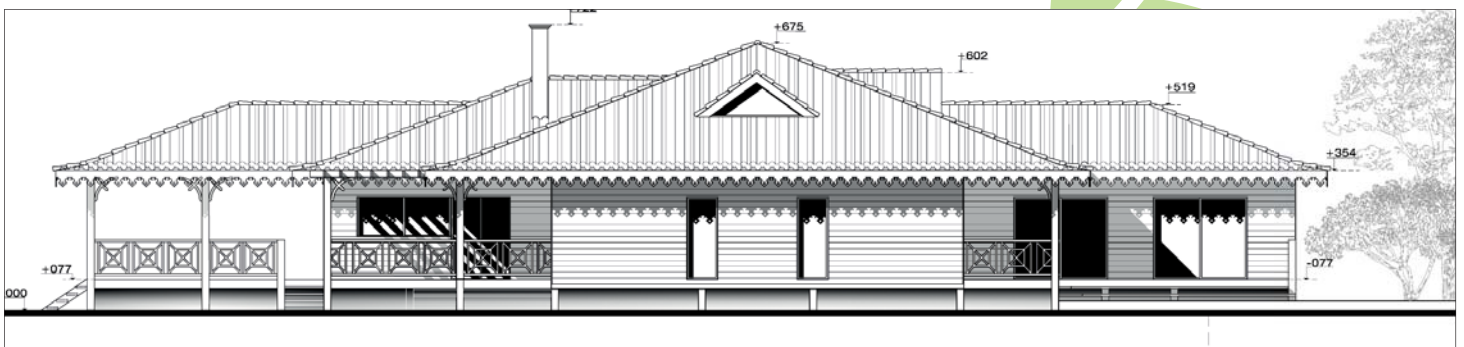
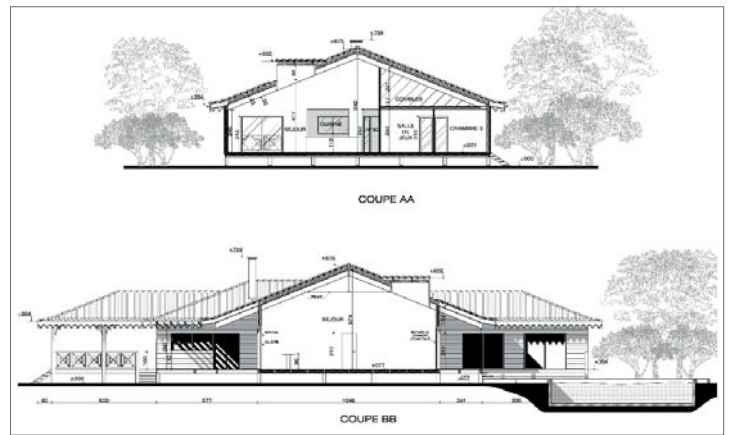
3D model of 2 timber-framed houses in Arcachon

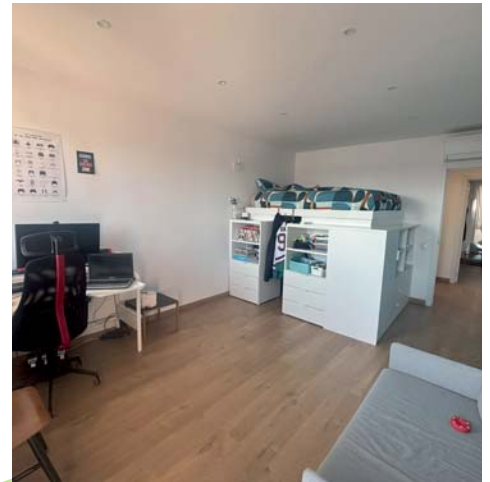
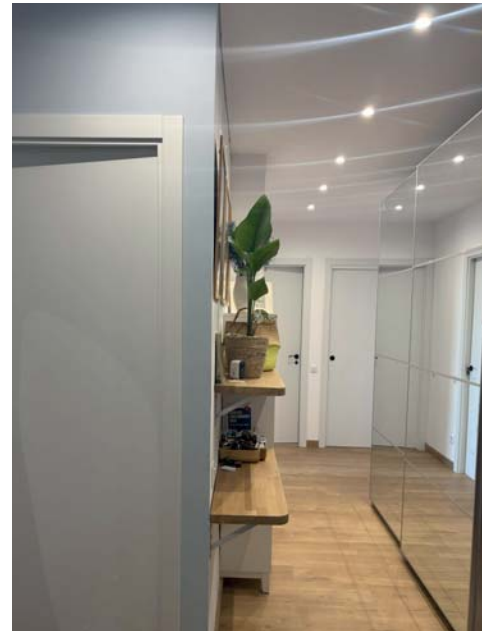
Parenteau Ithurbide Agency
2006-2007



Plans, sections, and main façades of a timber-framed house

Parenteau Ithurbide Agency
2006





Complete renovation of a 135 m² duplex apartment in the Glories district of Barcelona, comprising a master suite, 3 single bedrooms, a laundry room, 2 bathrooms, and a living area open to the terrace and surrounding the Glories Plaza.

